

SIGNALS AFTER THE NOISE 2

Hunt 06 // Post-Intrusion Investigation

LAW-Cyber-Range Sentinel • azwks-phtg-01 • 2025-12-13 09:00–18:00 UTC

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CONFIDENTIAL // CYBER RANGE OPERATIONS

EXECUTIVE SUMMARY

Incident Overview

On 13 December 2025, a threat operator achieved access to the PHTG HealthCloud estate via password reuse on account vmadminusername and conducted a structured post-intrusion campaign against workstation azwks-phtg-01. The operator laterally moved via RDP from azwks-phtg-02 (10.0.0.152) to azwks-phtg-01 (10.0.0.105) at 09:48 UTC.

Attack Chain Summary

Within minutes of landing, the operator executed master staging script `_ps1` which applied a temporary Defender exclusion, deployed the primary implant (bitsadmin.exe renamed as PHTGHealthCloudSvc.exe), established scheduled task and Run key persistence, registered a custom event log source, and stood up dual C2 beacons to health-cloud.cc infrastructure. AMSI was probed, process lineage was deliberately broken via `cmd.exe`, and at 10:14 UTC the operator opened a `PROCESS_ALL_ACCESS` handle on LSASS followed by confirmed memory read at 10:17 UTC indicating credential harvesting.

Key Findings

29 flags documented across 8 phases. Initial access via password reuse (T1078). Lateral movement via RDP `mstsc.exe`. Dual C2 channels: SvcExe healthcheck loop + encoded PowerShell beacons to `updates.health-cloud.cc` and `status.health-cloud.cc`. Persistence via scheduled tasks, Run key (PHTGHealthCloudTray), and startup LNK. Defender operating in detect-only mode — persistence landed unblocked. LSASS credential dump confirmed via `ReadProcessMemoryApiCall` at 10:17 UTC.

Risk Rating

CRITICAL. Operator achieved persistence, redundant C2, and credential access. No active blocking by Defender during the intrusion window. Immediate actions required: isolate azwks-phtg-01, rotate all vmadminusername credentials, remove Defender exclusions, and block health-cloud.cc at perimeter.

ATTACK TIMELINE — ALL FLAGS WITH EXACT ANSWERS

Phase 00 — Mission Brief

Q00 Gate Phrase	
Question	Acknowledge readiness with the phrase from the Hub.
Answer	CYBER RANGE//0x48554E54 (0x48=H 0x55=U 0x4E=N 0x54=T)
MITRE ATT&CK	—

Phase 01 — Cold Trail

Q01 Initial Access Vector		09:48 UTC
Question	Hunt 03 logged failed logons from several regions before the successful one. Test that assumption. If the successful access was not brute force, what was the actual vector?	
Answer	password reuse	
MITRE ATT&CK	T1078 — Valid Accounts	
KQL	<pre>SigninLogs where TimeGenerated between (datetime(2025-12-13T09:00:00Z) .. datetime(2025-12-13T18:00:00Z)) where ResultType == 0 project TimeGenerated, UserPrincipalName, IPAddress, AuthenticationRequirement order by TimeGenerated asc</pre>	

Q02 Lateral Movement Logon		09:48 UTC
Question	Where did the operator land? Name the account, source IP, and target host.	
Answer	vmadminusername, 10.0.0.152, azwks-phtg-01	
MITRE ATT&CK	T1021.001 — Remote Desktop Protocol	
KQL	<pre>DeviceNetworkEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where RemotePort == 3389 project TimeGenerated, DeviceName, LocalIP, RemoteIP, InitiatingProcessAccountName, InitiatingProcessFileName order by TimeGenerated asc</pre>	

Q03 Onward Movement Check		09:48 UTC
Question	Did the operator pivot onward from phtg-01? Use the secondary host's IP as source, check DeviceLogonEvents across the fleet.	
Answer	None	
MITRE ATT&CK	T1021 — Remote Services	
KQL	<pre>DeviceLogonEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where RemoteIP == "10.0.0.105" where ActionType == "LogonSuccess" project TimeGenerated, DeviceName, AccountName, RemoteIP</pre>	

Phase 02 — First Footsteps

Q04 First Operator Script		10:11:42
Question	After lateral movement, what is the first script the operator launched under their own account context? Full path.	
Answer	C:\Users\vmAdminUsername\Documents\PHTG_.ps1	
MITRE ATT&CK	T1059.001 — PowerShell	
KQL	<pre>DeviceProcessEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where DeviceName == "azwks-phtg-01" where AccountName == "vmadminusername" where ProcessCommandLine contains ".ps1" project TimeGenerated, AccountName, ProcessCommandLine, InitiatingProcessFileName order by TimeGenerated asc</pre>	

Q05 Operator Concealment Flags		10:11:42
Question	Look at the command line that invoked the Q04 script. Two PowerShell flags signal operator intent. Name both with their values.	
Answer	-WindowStyle Hidden -ExecutionPolicy Bypass	
MITRE ATT&CK	T1564.003 — Hide Artifacts: Hidden Window	

Q06 Operator Tooling Workspace 10:11:42	
Question	Where did the operator stage their tooling? The root directory under ProgramData, three subdirectories sit underneath it.
Answer	C:\ProgramData\PHTG\HealthCloud
MITRE ATT&CK	T1074.001 — Local Data Staging
KQL	<pre>DeviceFileEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where DeviceName == "azwks-phtg-01" where FolderPath contains "ProgramData\PHTG" summarize by FolderPath order by FolderPath asc</pre>
Q07 Scheduled Task Name 10:11:42	
Question	What is the name of the first scheduled task the operator created?
Answer	PHTG HealthCloud Compliance FLAG-01
MITRE ATT&CK	T1053.005 — Scheduled Task/Job
KQL	<pre>DeviceEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ActionType == "ScheduledTaskCreated" where DeviceName == "azwks-phtg-01" project TimeGenerated, AdditionalFields order by TimeGenerated asc</pre>
Q08 Masquerading Binary — LOLBin Identification 10:12:17	
Question	One process is not what it claims. What binary was renamed and running as PHTGHealthCloudSvc.exe? Explain how it was isolated from noise.
Answer	PHTGHealthCloudSvc.exe is a renamed bitsadmin.exe (LOLBin); isolated because it ran from operator staging path under ProgramData rather than System32
MITRE ATT&CK	T1036.003 — Masquerading: Rename System Utilities
KQL	<pre>DeviceProcessEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where DeviceName == "azwks-phtg-01" where FileName != OriginalFileName project TimeGenerated, FileName, OriginalFileName, FolderPath, AccountName order by TimeGenerated asc</pre>

Phase 03 — Quiet Roots

Q09 Registry Activity Volume 09:48--	
Question	Volume check. How many registry modification events fired under vmadminusername on phtg-01 AFTER lateral movement?
Answer	280
MITRE ATT&CK	T1112 — Modify Registry
KQL	<pre>DeviceRegistryEvents where TimeGenerated >= datetime(2025-12-13T09:48:00Z) where DeviceName contains "phtg-01" where InitiatingProcessAccountName =~ "vmadminusername" count</pre>
Q10 Persistence Signal Isolation 10:12:59	
Question	280 events is mostly noise. Which registry path actually matters for persistence?
Answer	HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
MITRE ATT&CK	T1547.001 — Registry Run Keys / Startup Folder
KQL	<pre>DeviceRegistryEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where DeviceName == "azwks-phtg-01" where InitiatingProcessAccountName =~ "vmadminusername" where RegistryKey contains "CurrentVersion\\Run" project TimeGenerated, ActionType, RegistryKey, RegistryValueName, RegistryValueData</pre>
Q11 Run Key Value Name 10:12:59	
Question	The Run key carries multiple values. Edge auto-launch is legitimate. Which value name points to the operator's tooling?
Answer	PHTGHealthCloudTray
MITRE ATT&CK	T1547.001 — Registry Run Keys / Startup Folder
Q12 Custom Event Log Source 10:11:43	
Question	What custom Application event log source did the operator register?
Answer	PHTGHealthCloud
MITRE ATT&CK	T1562.002 — Disable Windows Event Logging
KQL	<pre>DeviceRegistryEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where RegistryKey contains "EventLog\\Application" where InitiatingProcessAccountName =~ "vmadminusername" project TimeGenerated, RegistryKey, RegistryValueName</pre>

Q13 Beacon Pair — Decode		10:12:16–10:13:44
Question	Alongside the SvcExe healthcheck loop, the operator fired two encoded PowerShell beacons. Decode both. What endpoints did they contact? Report both in chronological order with the parent domain.	
Answer	The first beacon contacted https://status.health-cloud.cc/api/checkin and the second contacted https://status.health-cloud.cc/api/status , both under the parent domain health-cloud.cc	
MITRE ATT&CK	T1071.001 — Application Layer Protocol: Web / T1132.001 — Data Encoding: Standard Encoding	
KQL	<pre>DeviceEvents where TimeGenerated between (datetime(2025-12-13T10:12:00Z) .. datetime(2025-12-13T10:14:00Z)) where ActionType == "PowerShellCommand" where AdditionalFields contains "[Convert]::FromBase64String" project TimeGenerated, AdditionalFields</pre>	

Q14 Third Persistence Mechanism		10:11:43
Question	The operator made a system-level (HKLM) registry change on phtg-01. It's not persistence in the classic sense — it gives their tooling a different capability. Identify the key path.	
Answer	HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Services\EventLog\Application\PHTGHealthCloud	
MITRE ATT&CK	T1562.002 — Disable Windows Event Logging	
KQL	<pre>DeviceRegistryEvents where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where DeviceName == "azwks-phtg-01" where RegistryKey contains "HKEY_LOCAL_MACHINE" and RegistryKey contains "EventLog" where InitiatingProcessAccountName =~ "vmadminusername" project TimeGenerated, RegistryKey, RegistryValueName, RegistryValueData</pre>	

Phase 04 — The Beacon Pair

Q15 Healthcheck Execution Count		10:12:17–18:00
Question	The masquerade binary from Q08 runs in a loop, beaconing with /healthcheck at regular short intervals. How many healthcheck executions fired during post-access activity?	
Answer	22	
MITRE ATT&CK	T1071.001 — Application Layer Protocol: Web / T1036.003 — Masquerading	
KQL	<pre>DeviceProcessEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ProcessCommandLine contains "/healthcheck" count</pre>	

Q16 Encoded Beacon Endpoints		10:12:16–10:13:44
Question	Alongside the SvcExe healthcheck loop, the operator fired two encoded PowerShell beacons. Decode both. What endpoints did they contact? Report both in chronological order.	
Answer	The first beacon contacted <code>https://status.health-cloud.cc/api/checkin</code> and the second contacted <code>https://status.health-cloud.cc/api/status</code> , both under the parent domain <code>health-cloud.cc</code>	
MITRE ATT&CK	T1071.001 — Application Layer Protocol: Web / T1132.001 — Data Encoding: Standard Encoding	

Q17 Two Beacons, Why?		—
Question	The operator runs TWO beaconding mechanisms in parallel. What is the operational benefit of running both?	
Answer	Check-in then deploy pattern: if one channel is spotted, the operator can abandon it while the second maintains access. If one is cut off, the second ensures C2 continuity without interruption. (T1090 fallback/redundancy)	
MITRE ATT&CK	T1090 — Proxy / Redundant C2	

Q18 Deployment Pattern Recognition		10:12:16–10:12:17
Question	An outbound connection fires at 10:12:16. PHTGHealthCloudSvc.exe launches at 10:12:17. A 1-second gap. What deployment pattern is the operator using?	
Answer	Check-in then deploy. <code>_ps1</code> checks in with C2 via an outbound call, and the completed check-in triggers PHTGHealthCloudSvc.exe to deploy one second later.	
MITRE ATT&CK	T1105 — Ingress Tool Transfer	

Phase 05 — Outbound Whispers

Q19 Operator Outbound Domains		10:12:16 / 10:13:44
Question	What domains did the operator's PowerShell reach during post-access activity? List both in chronological order.	
Answer	<code>updates.health-cloud.cc</code> , <code>status.health-cloud.cc</code>	
MITRE ATT&CK	T1071.001 — Application Layer Protocol: Web	
KQL	<pre>DeviceNetworkEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ActionType == "ConnectionSuccess" where RemoteUrl endswith ".cc" project TimeGenerated, RemoteUrl, RemoteIP order by TimeGenerated asc</pre>	

Q20 AMSI Probe Identification 10:14:10	
Question	After outbound succeeded, the operator ran a plain PowerShell script from their staging Bin directory. Name the script and explain what it is doing.
Answer	amsi_probe.ps1 — tests whether the Antimalware Scan Interface is active and responsive on the host. The operator runs it to check if AMSI will flag subsequent activity, allowing them to gauge defensive coverage before deploying more aggressive tooling.
MITRE ATT&CK	T1562.001 — Impair Defenses: Disable or Modify Tools
KQL	<pre>DeviceProcessEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T10:12:00Z) .. datetime(2025-12-13T18:00:00Z)) where ProcessCommandLine contains "Bin" and ProcessCommandLine contains ".ps1" where ProcessCommandLine !contains "task_FLAG" project TimeGenerated, ProcessCommandLine order by TimeGenerated asc</pre>

Q21 Lineage Break Pattern 10:15:15 / 10:16:09	
Question	The operator twice launched follow-on payloads via cmd.exe instead of running them directly. Identify both invocations and explain why.
Answer	Invocation 1 (10:15:15): cmd.exe /c powershell.exe -NoProfile -ExecutionPolicy Bypass -File "C:\ProgramData\PHTG\HealthCloud\Bin\hc_lineage.ps1" Invocation 2 (10:16:09): cmd.exe /c "C:\ProgramData\PHTG\HealthCloud\phtg_health_diag_update_FLAG-22.bat" Using cmd.exe breaks the parent process lineage, inserting a shell between the operator's PowerShell session and the payload to obscure the spawn chain from detections.
MITRE ATT&CK	T1059.003 — Windows Command Shell
KQL	<pre>DeviceProcessEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T10:00:00Z) .. datetime(2025-12-13T11:30:00Z)) where FileName == "cmd.exe" where ProcessCommandLine !contains "whoami" project TimeGenerated, ProcessCommandLine, InitiatingProcessCommandLine order by TimeGenerated asc</pre>

Phase 06 — Doors Held Open

Q22 Defender Tampering		10:11:42 / 10:12:18 / 10:14:30
Question	The operator made Defender quieter after persistence landed. What did they exclude? One is a path, one is a process.	
Answer	Path: C:\ProgramData\PHTG\HealthCloud\Cache Process: C:\ProgramData\PHTG\HealthCloud\PHTGHealthCloudSvc.exe	
MITRE ATT&CK	T1562.001 — Impair Defenses: Disable or Modify Tools	
KQL	<pre>DeviceRegistryEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where RegistryKey contains "Defender" and RegistryKey contains "Exclusion" project TimeGenerated, RegistryKey, RegistryValueName, RegistryValueData order by TimeGenerated asc</pre>	
Q23 Defender Detection Outcome		10:13:14–10:13:16
Question	Defender generated two AntivirusReport events on the PHTG HealthCloud.lnk artefact. Did it block the persistence? What does WasExecutingWhileDetected tell you?	
Answer	Defender detected PHTG HealthCloud.lnk but did not block it. WasExecutingWhileDetected being false means the artefact had already executed and established persistence before Defender detected it, revealing a defensive posture that is too slow to prevent execution.	
MITRE ATT&CK	T1562.001 — Impair Defenses	
KQL	<pre>DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ActionType == "AntivirusReport" where FileName contains "HealthCloud.lnk" project TimeGenerated, FileName, FolderPath, AdditionalFields</pre>	
Q24 Temporary Defender Exclusion		10:11:42
Question	Inside _.ps1 itself the operator applied an exclusion then removed it within seconds. Identify the path, prove the add-then-remove pattern, and explain why.	
Answer	Path temporarily excluded: C:\Users\vmAdminUsername\Documents\PHTG Pattern: Single PowerShell command — Add-MpPreference adds exclusion, _.ps1 executes, then Remove-MpPreference removes it immediately after. Reason: Minimises the detection window so the path does not remain excluded long enough to be noticed by a defender reviewing Defender's configuration.	
MITRE ATT&CK	T1562.001 — Impair Defenses	
KQL	<pre>DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T10:11:00Z) .. datetime(2025-12-13T10:14:00Z)) where ActionType == "PowerShellCommand" where AdditionalFields contains "MpPreference" project TimeGenerated, AdditionalFields</pre>	

Q25 Startup Execution Validation		10:13:04–10:13:08
Question	How many times did the HealthCloudTray.ps1 startup command execute during the investigation window?	
Answer	2	
MITRE ATT&CK	T1547.001 — Boot/Logon Autostart: Registry Run Keys	
KQL	<pre>DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where AdditionalFields contains "HealthCloudTray" count</pre>	

Q26 Custom Event Log Source Purpose		10:11:43
Question	What does registering the custom Application event log source enable for the operator's tooling?	
Answer	Registering the source lets the operator's tooling call Write-EventLog to write entries to the Application log, blending C2 activity into trusted system telemetry where it appears as legitimate application logging and avoids file-based detection.	
MITRE ATT&CK	T1562.002 — Disable Windows Event Logging	

Phase 07 — Hands on the Vault

Q27 LSASS Access Anomaly		10:14:37–10:14:38
Question	139 OpenProcessApiCall events target lsass.exe. Most are baseline. One is not. Name the initiating process and account context that is not system.	
Answer	vmadminusername, powershell.exe	
MITRE ATT&CK	T1003.001 — OS Credential Dumping: LSASS Memory	
KQL	<pre>DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ActionType == "OpenProcessApiCall" where FileName =~ "lsass.exe" project TimeGenerated, AccountName, InitiatingProcessFileName, InitiatingProcessAccountName, AdditionalFields order by TimeGenerated asc</pre>	

Q28 Access Right Escalation 10:14:37–10:14:38	
Question	The anomalous LSASS access fired with two DesiredAccess values one second apart. Which one grants full access and why is the escalation significant?
Answer	2047999 (0x1F3FFF — PROCESS_ALL_ACCESS equivalent) Escalation: First request (5136) was limited access; one second later escalated to full process access — indicating deliberate privilege escalation to enable memory read/dump.
MITRE ATT&CK	T1003.001 — OS Credential Dumping
KQL	<pre> DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T10:14:00Z) .. datetime(2025-12-13T10:15:00Z)) where ActionType == "OpenProcessApiCall" where FileName =~ "lsass.exe" where InitiatingProcessAccountName =~ "vmadminusername" project TimeGenerated, AdditionalFields order by TimeGenerated asc </pre>
Q29 Credential Dump Confirmation 10:17:36	
Question	Opening a full-access handle to LSASS is not dumping yet. What ActionType would you expect if the operator actually read LSASS memory? Confirm it fired on phtg-01.
Answer	ReadProcessMemoryApiCall Confirmed: fired at 10:17:36 on lsass.exe initiated by powershell.exe under vmadminusername
MITRE ATT&CK	T1003.001 — OS Credential Dumping: LSASS Memory
KQL	<pre> DeviceEvents where DeviceName == "azwks-phtg-01" where TimeGenerated between (datetime(2025-12-13T09:48:00Z) .. datetime(2025-12-13T18:00:00Z)) where ActionType == "ReadProcessMemoryApiCall" where InitiatingProcessFileName =~ "powershell.exe" where FileName =~ "lsass.exe" project TimeGenerated, ActionType, FileName, InitiatingProcessFileName </pre>

MITRE ATT&CK; TECHNIQUE SUMMARY

Technique ID	Technique Name	Tactic	Flags
T1078	Valid Accounts	Initial Access / Defense Evasion	Q01
T1021.001	Remote Desktop Protocol	Lateral Movement	Q02
T1021	Remote Services	Lateral Movement	Q03
T1059.001	PowerShell	Execution	Q04, Q05
T1564.003	Hide Artifacts: Hidden Window	Defense Evasion	Q05
T1074.001	Local Data Staging	Collection	Q06
T1053.005	Scheduled Task/Job	Persistence / Execution	Q07
T1036.003	Masquerading: Rename System Utilities	Defense Evasion	Q08
T1112	Modify Registry	Defense Evasion	Q09
T1547.001	Registry Run Keys / Startup Folder	Persistence	Q10, Q11, Q25
T1562.002	Disable Windows Event Logging	Defense Evasion	Q12, Q26
T1071.001	Application Layer Protocol: Web	C2	Q13–Q16, Q19
T1090	Proxy / Redundant C2	C2	Q17
T1105	Ingress Tool Transfer	C2	Q18
T1562.001	Impair Defenses: Disable/Modify Tools	Defense Evasion	Q20, Q22, Q23, Q24
T1059.003	Windows Command Shell	Execution	Q21
T1003.001	OS Credential Dumping: LSASS Memory	Credential Access	Q27, Q28, Q29

INDICATORS OF COMPROMISE

Type	Indicator	Context
Domain	updates.health-cloud.cc	C2 payload delivery — first contact 10:12:16
Domain	status.health-cloud.cc	C2 beacon checkin/status — 10:13:44
URL	https://status.health-cloud.cc/api/checkin	FLAG-09 beacon endpoint
URL	https://status.health-cloud.cc/api/status	FLAG-10 beacon endpoint
File	PHTGHealthCloudSvc.exe	Renamed bitsadmin.exe (LOLBin) — staging path
File	_.ps1	Master staging/deployment script
File	amsi_probe.ps1	AMSI evasion detection probe
File	hc_lineage.ps1	Lineage-breaking payload
File	PHTG HealthCloud.lnk	Startup persistence LNK — Defender detected, not blocked
Path	C:\ProgramData\PHTG\HealthCloud\	Operator staging root (Cache, Bin, subdirs)

Type	Indicator	Context
Path	C:\Users\vmAdminUsername\Documents\PHTG	Initial staging / temporary exclusion path
Registry	HKCU\...\CurrentVersion\Run: PHTGHealthCloudTray	Run key persistence — survives logon
Event Log	Application: PHTGHealthCloud (custom source)	Operator log blending mechanism
Account	vmadminusername	Compromised account — RDP lateral movement + LSASS access

REMIEDIATION RECOMMENDATIONS

IMMEDIATE — Containment (Do Now)

Isolate azwks-phtg-01	Immediately remove the host from the network. The operator's implant (PHTGHealthCloudSvc.exe) and dual C2 channels may still be active. Do not power off — preserve volatile memory for forensic analysis.
Block health-cloud.cc at perimeter	Add DNS and firewall blocks for health-cloud.cc, status.health-cloud.cc, and updates.health-cloud.cc across all egress points. Check proxy logs for any other hosts that may have contacted these domains.
Rotate vmadminusername credentials estate-wide	The compromised password was reused — assume it may be valid on other systems. Force immediate password reset for vmadminusername and audit all systems where this account has authenticated in the past 30 days.
Revoke active sessions	Terminate all active RDP and remote sessions for vmadminusername. Review azwks-phtg-02 (the source of lateral movement) for any additional operator tooling or staging directories.

SHORT TERM — Persistence Removal (Within 24 Hours)

Remove all three persistence mechanisms	Delete scheduled tasks matching 'PHTG HealthCloud Compliance*'. Remove Run key value PHTGHealthCloudTray from HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run. Delete PHTG HealthCloud.lnk from the Startup folder.
Remove Defender exclusions	Delete Defender exclusions for C:\ProgramData\PHTG\HealthCloud\Cache and PHTGHealthCloudSvc.exe. Run a full scan of the host after exclusions are removed to surface any additional malicious files.
Remove custom event log source	Delete the registry key HKLM\SYSTEM\ControlSet001\Services\EventLog\Application\PHTGHealthCloud. Review the Application event log for any entries written by this source during the intrusion window.
Remove operator staging directories	Delete C:\ProgramData\PHTG\HealthCloud\ and C:\Users\vmAdminUsername\Documents\PHTG\ in their entirety after forensic imaging. Hash and submit all binaries found for threat intelligence.
Assume credentials are compromised	ReadProcessMemoryApiCall on LSASS at 10:17 UTC confirms credential harvesting. Treat all credentials cached on azwks-phtg-01 as compromised — rotate service accounts, local admin passwords, and any domain credentials that may have been logged on to this host.

MEDIUM TERM — Detection & Hardening (Within 1 Week)

Enforce Defender active remediation	Defender was in detect-only mode — WasExecutingWhileDetected=false confirms no active blocking occurred. Review and update Defender policy to block on detection. Enable Tamper Protection to prevent future exclusion modifications via registry.
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Enable LSASS protection	Enable Protected Process Light (PPL) for LSASS via registry: HKLM\SYSTEM\CurrentControlSet\Control\Lsa — RunAsPPL = 1. This prevents non-PPL processes from opening high-privilege handles to LSASS. Also enable Credential Guard where possible.
Block LOLBin abuse	PHTGHealthCloudSvc.exe was a renamed bitsadmin.exe. Implement AppLocker or WDAC rules to prevent execution of known LOLBins (bitsadmin.exe, certutil.exe, etc.) from non-standard paths. Alert on OriginalFileName mismatches in EDR telemetry.
Alert on Defender exclusion modifications	Create detection rules for registry writes to HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions* by non-SYSTEM processes. Temporary exclusion add-then-remove within seconds is a strong indicator of deliberate evasion.
Detect encoded PowerShell beacons	Create rules for PowerShell commands containing [Convert]::FromBase64String followed by Invoke-WebRequest or Invoke-Expression. Alert on PowerShell processes making outbound connections to low-reputation or newly registered domains.
Monitor for LSASS handle escalation	Alert on OpenProcessApiCall events targeting lsass.exe with DesiredAccess values above 0x1000 from non-system processes. ReadProcessMemoryApiCall on lsass.exe from any non-EDR process should trigger immediate response.
Audit scheduled task creation	Alert on ScheduledTaskCreated events where the task action runs PowerShell with -WindowStyle Hidden or -ExecutionPolicy Bypass. Baseline legitimate scheduled tasks and alert on deviations.

LONG TERM — Strategic Improvements (Within 30 Days)

Eliminate password reuse	The initial access vector was password reuse — a previously compromised password worked without any MFA challenge. Enforce unique passwords via enterprise password manager, implement MFA for all RDP and remote access, and consider Privileged Access Workstations (PAWs) for admin accounts.
Network segmentation	The operator moved laterally via RDP from azwks-phtg-02 to azwks-phtg-01 without restriction. Implement micro-segmentation to restrict RDP access between workstations. Workstation-to-workstation RDP should be blocked by default and only permitted via jump hosts.
Improve EDR coverage and tuning	The operator's entire intrusion — from RDP landing to LSASS dump — completed with no EDR-triggered response. Review EDR policy configuration, ensure all detections route to a monitored queue, and establish SLAs for alert triage. Consider enabling automatic isolation on high-confidence detections.
Hunt for additional victims	The operator's tooling (PHTGHealthCloudSvc.exe, amsi_probe.ps1, hc_lineage.ps1) and C2 infrastructure (health-cloud.cc) should be hunted across the entire estate. Run the KQL queries from this report against all devices in the Sentinel workspace to identify any additional compromised hosts.
Threat intelligence sharing	Submit health-cloud.cc infrastructure, PHTGHealthCloudSvc.exe hashes, and operator TTPs to your threat intelligence platform. The structured, multi-phase tradecraft suggests an experienced operator — cross-reference with known threat actor profiles.

